Hai Duong's
Curriculum Vitae

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1 Education

- **PhD**, Computer Science, George Mason University, Fairfax, VA 2022–present

  Advisor: ThanhVu Nguyen

- **MS**, Electrical Engineering, Hanoi University of Science and Technology, Hanoi, VN 2019–2021

  Advisor: Quoc-Cuong Nguyen

- **BS**, Electrical Engineering, Hanoi University of Science and Technology, Hanoi, VN 2014–2019

2 Experience

- **Applied Scientist Intern**, Amazon Web Services, Amazon 2026

  Developed SMT-based path enumeration and fuzzing for AWS compliance, achieving 100% test coverage and exposing legacy-migration divergences.

  Built automated Java/Python to ControlsDSL migration framework with bidirectional equivalence verification, porting 40% of staging rules to production.

- **Research Assistant**, ROARS Lab, George Mason University 2022–present

  Developed a verification tool, NeuralSAT (P6, S2), ranked 2nd in VNN-COMPs.

  Developed optimization techniques for DNN verification, e.g., neuron stabilization (P7), network composition (P4), DRL-based branching heuristic (P1), generating and checking unsat proofs (P5), structural perturbations (P3), dual perturbations (P2), training for verification (P8).

  Developed abtract domains for DNN verification, e.g., constrained inputs (U1), transformers-based models (U2), multi-parameter bit-flip perturbations (U3)

- **Research Assistant**, BachLe's Lab, University of Melbourne 2021–2022

  Worked on a graph-based source code modeling and explanation technique (P9).

- **Research Assistant**, Sensor Lab, Hanoi University of Science and Technology 2019–2021

  Developed a speech enhancement system using graph-based neural beamforming (P11).

  Developed a small-footprint keyword spotting system using deformable convolution (P10).

3 Awards and Honors

- Research Grants ($5,000 API Credits), Thinking Machines 2026
- Silver Reviewer Award, ICML 2026
- Outstanding PhD Student Award, GMU 2026
- Highlight Paper Award [P2], CVPR 2026
- Broadening Participation Scholarship Award, CVPR 2026

- Spotlight Paper Award [P4], NeurIPS 2025
- Scholar Award, NeurIPS 2025
- Graduate Travel Award, GMU 2025
- NeuralSAT ranked 2nd, VNN-COMP 2024, 2025
- NeuralSAT ranked 4th, VNN-COMP 2023, 2026
- New Participant Award, VNN-COMP 2023
- Graduate Scholarship Award, HUST 2019–2021
- Outstanding Undergraduate Award, HUST 2015–2018

4 Publications

4.1 Refereed Conference/Journal Papers (Full Papers)

P1 Hai Duong, Thanh Le, and ThanhVu Nguyen. "Verifying Neural Networks with Reinforcement Learning". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2026, to appear

P2 Hai Duong, Lam Nguyen, Thanh Le, and ThanhVu Nguyen. "Verifying Neural Network Robustness with Dual Perturbations". In: *Computer Vision and Pattern Recognition (CVPR)*. 2026, pages 27916–27925

Highlight Paper

P3 Hai Duong, Thanh Le, Lam Nguyen, and ThanhVu Nguyen. "Verifying structural robustness of deep neural network". In: *Proceedings of the ACM on Software Engineering* 3.Foundations of Software Engineering (FSE) (2026), pages 1492–1514

P4 Hai Duong, David Shriver, ThanhVu Nguyen, and Matthew Dwyer. "Compositional neural network verification via assume-guarantee reasoning". In: *Advances in Neural Information Processing Systems (NeurIPS)* 38 (2025), pages 64158–64182

Spotlight Paper

P5 Hai Duong, ThanhVu Nguyen, and Matthew Dwyer. "Generating and checking dnn verification proofs". In: *Advances in Neural Information Processing Systems (NeurIPS)* 38 (2025), pages 65887–65909

P6 Hai Duong, ThanhVu Nguyen, and Matthew B Dwyer. "Neuralsat: A high-performance verification tool for deep neural networks". In: *Computer Aided Verification (CAV)*. Springer. 2025, pages 409–423

P7 Hai Duong, Dong Xu, ThanhVu Nguyen, and Matthew B Dwyer. "Harnessing neuron stability to improve dnn verification". In: *Proceedings of the ACM on Software Engineering* 1.Foundations of Software Engineering (FSE) (2024), pages 859–881

P8 Dong Xu, Nusrat Jahan Mozumder, Hai Duong, and Matthew B Dwyer. "Training for verification: Increasing neuron stability to scale dnn verification". In: *International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*. Springer. 2024, pages 24–44

P9 Thanh-Dat Nguyen, Thanh Le-Cong, Duc-Minh Luong, Hai Duong, Xuan-Bach D Le, David Lo, and Quyet-Thang Huynh. "Ffl: Fine-grained fault localization for student programs via syntactic and semantic reasoning". In: *2022 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. IEEE. 2022, pages 151–162

P10 Huu Binh Nguyen, Hai Duong, Anh Xuan Tran Thi, and Quoc Cuong Nguyen. "Efficient keyword spotting system using deformable convolutional network". In: *IETE Journal of Research* 69.7 (2023), pages 4196–4204

P11 Huu Binh Nguyen, Hai Duong, Tien Dat Bui, Hoang Ngoc Chau, and Quoc Cuong Nguyen. "Multi-channel speech enhancement using a minimum variance distortionless response beamformer based on graph convolutional network". In: *International Journal of Advanced Computer Science and Applications* 13.10 (2022)

4.2 Refereed Conference/Workshop Papers (Short Papers)

S1 Cruise Song, Hai Duong, ThanhVu Nguyen, and Vijay Ganesh. "LeanVIPR: A Formally Verified Certificate Checker for Exact MIP (Workshop)". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2026, to appear

S2 Hai Duong and ThanhVu Nguyen. "Neuralsat: Scaling constraint solving for dnn verification (competition contribution)". In: *International Symposium on AI Verification (SAIV)*. Springer. 2025, pages 253–259

S3 ThanhVu Nguyen, KimHao Nguyen, and Hai Duong. "SymInfer: Inferring Numerical Invariants using Symbolic States". In: *International Conference on Software Engineering (ICSE)*. IEEE, 2022, pages 197–201

4.3 Unpublished

U1 Hai Duong, Thanh Le, and ThanhVu Nguyen. "CLAD: Constrained Abstract Domain for Neural Network Verification". In: *Submitted*. 2026

U2 Hai Duong, Thanh Le, and ThanhVu Nguyen. "ZonoGPT: Towards An Abstract Domain for Verifying Large GPT Models". In: *Submitted*. 2026

U3 Hai Duong, Thanh Le, Nguyen Ho, and ThanhVu Nguyen. "Towards Verifying Neural Networks Against Multi-Parameter Bit-Flip Perturbations". In: *Submitted*. 2026

U4 Thanh Le, Hai Duong, ThanhVu Nguyen, and Takeshi Matsumura. "Verca: Verifying Robustness of Constrained-Latent Autoencoders". In: *Submitted*. 2026

U5 Junyu Yin, Lingda Li, Hai Duong, Adolfy Hoisie, and Keren Zhou. "Trace-Driven DL-based Framework for GPU Performance Modeling". In: *Submitted*. 2026

U6 Thanh Le, Hai Duong, ThanhVu Nguyen, and Takeshi Matsumura. "Formal Verification of DNN-based Semantic Communication to Adversarial Noise". In: *Submitted*. 2026

U7 Thanh Le, Hai Duong, Yusheng Ji, ThanhVu Nguyen, and John C.S. Lui. "FGGM: Formal Grey-box Gradient Method for Attacking DRL-based MU-MIMO Scheduler". In: *arXiv*. 2025

U8 Hai Duong, ThanhVu Nguyen, and Matthew Dwyer. "A DPLL(T) Framework for Verifying Deep Neural Networks". In: *arXiv*. 2024

5 Miscellaneous

- Contribute to proposals:
  - Amazon Research Award (Build on Trainium: Responsible AI): Leveraging AWS Trainium for Verifiable AI and ML-Assisted Mathematical Reasoning (2026). $250,000 Training Credits. Amazon.
  - NVIDIA Academic Grant Program: Trustworthy AI: Bringing Scalability and Assurance to DNN Verification (2025). DGX Spark System. NVIDIA.
  - Amazon Research Award (Automated Reasoning): Scalable and Precise DNN Constraint Solving with Abstraction and Conflict Clause Learning (2023). $50,000 unrestricted gift. Amazon.
  - NSF CAREER: NeuralSAT: A Constraint-Solving Framework for Verifying Deep Neural Networks (2023). NSF 2238133. 8/1/2023–7/31/2028, $510,509. NSF.
- Help review conference papers:
  - ISSTA'23
  - PLDI'24, OOPSLA'24
  - ICLR'25
  - CVPR'26, ICML'26, ECCV'26, NeurIPS'26, AAAI'26
  - POPL'27, ICLR'27